METRICMIND

Agentic Semantic BI Engine

Project Report

Submitted by: Utukuri Geetha Sai Ganesha Bebi Malathi

Project Domain: Enterprise Analytics & Conversational BI

Prototype implementation using sample demonstration data

Abstract

MetricMind is a conversational business intelligence (BI) prototype designed to let users ask business-metric questions in natural language and receive structured, readable responses. The project explores how a governed analytics experience can make business metrics easier to access while presenting transparency information such as response metadata and representative SQL. The current demonstration uses a small pandas DataFrame containing quarterly revenue and cost values for Europe. It supports selected revenue, profit, profit-margin, quarter-comparison, and region-related questions. A FastAPI backend exposes a query endpoint, while a Retool interface provides the conversational dashboard and transparency panels. The current version is a prototype: calculations are performed in Python against sample data, and the displayed SQL is illustrative rather than executed against a live data warehouse.

1. Introduction

Business teams often need quick answers about revenue, costs, and margins but may not be comfortable writing queries or navigating analytical tools. Conversational BI offers a familiar question-and-answer interface. MetricMind demonstrates this idea through a natural-language interface, a Python-based calculation layer, and a structured API response.

The project is inspired by the need for consistent business definitions and transparent analytics. The prototype demonstrates selected rules and calculations; it should not be interpreted as a fully deployed enterprise semantic layer or a production database-connected system.

2. Problem Statement

Users may find it difficult to obtain consistent business metrics from raw data, especially when they must understand query syntax and metric definitions. Unvalidated or unclear query generation can also reduce trust in AI-assisted analytics. MetricMind addresses this problem at prototype level by mapping a defined set of questions to known calculations, returning structured answers, and exposing supporting metadata and representative SQL for inspection.

3. Objectives

- Provide a conversational interface for selected business questions.
- Calculate quarterly revenue, profit, and profit margin from a sample dataset.
- Support selected quarter-wise comparisons and revenue/cost summaries.
- Return structured API responses with answer, status, data-source label, and SQL text.
- Demonstrate frontend-to-backend integration using Retool and FastAPI.
- Handle unsupported questions without fabricating a business answer.

4. Scope and Limitations

Scope of the current prototype:

- Quarterly revenue, profit, and profit-margin calculations.
- Quarterly margin summaries and selected revenue/cost comparisons.

- A small demonstration dataset with Q1–Q4 values for Europe.
- A Retool interface that displays the answer and transparency information.

Current limitations:

- The data is synthetic/sample demonstration data, not company financial data.
- The displayed SQL is representative text and is not executed against a live database.
- Natural-language understanding is rule-based and supports a limited set of question patterns.
- Because the sample contains only Europe, it cannot support meaningful comparisons across multiple regions.
- The prototype does not yet establish production-grade semantic governance, authentication, audit controls, or live warehouse connectivity.

5. System Architecture

The prototype follows a simple request-and-response flow:

1. User enters a question in the Retool dashboard.
2. Retool sends the question to the FastAPI POST /query endpoint.
3. The Python function identifies supported question patterns and selects relevant sample rows.
4. Pandas performs the calculation for revenue, profit, or margin.
5. The backend returns a JSON response containing the answer, status, data-source label, and representative SQL.
6. Retool renders the governed-answer area and transparency panels.

Retool Conversational Dashboard
↓ HTTP POST /query
FastAPI Backend
↓ Rule-based question handling
Python + pandas sample-data calculations
↓ Structured JSON response
Retool Answer + Transparency Display

6. Technologies Used

Technology	Purpose in prototype
Python	Backend logic and metric calculations
pandas	Filtering, grouping, and aggregating sample data
FastAPI	HTTP API exposing the /query endpoint
Retool	Conversational dashboard and transparency interface
JSON	Structured request and response format
Jupyter Notebook	Development and testing environment
ngrok (if used during demo)	Temporary tunnel to expose the local API to Retool

7. Sample Dataset

The current prototype uses a small illustrative dataset with four quarterly records. All values below are demonstration values and should not be represented as actual company results.

Quarter	Region	Revenue	Cost	Profit
Q1	Europe	100,000	70,000	30,000

Q2	Europe	120,000	82,000	38,000
Q3	Europe	135,000	90,000	45,000
Q4	Europe	150,000	98,000	52,000

Profit = Revenue − Cost. Profit margin = (Profit ÷ Revenue) × 100. Using the sample records, quarterly profit margins are Q1: 30.00%, Q2: 31.67%, Q3: 33.33%, and Q4: 34.67%.

8. Main Features

- Conversational question input in Retool.
- Quarter-specific revenue calculations for Q1–Q4.
- Profit calculations based on revenue minus cost.
- Profit-margin calculation and quarter-wise margin summaries.
- Selected revenue and cost comparison summaries.
- Structured JSON response with success status, answer, data source, SQL text, and metadata.
- Unsupported-question response for questions outside the defined prototype rules.
- Transparency panels for inspecting the frontend request, API response, and displayed SQL text.

9. Testing and Results

The following behaviors were manually tested through the notebook and Retool interface. The figures are based on the sample dataset.

Test question / scenario	Expected sample result	Observed status
What was our Q1 revenue?	100,000	Passed in Retool
What was our Q2 revenue?	120,000	Passed in Retool
What was our Q3 revenue?	135,000	Passed in Retool
What was our Q4 revenue?	150,000	Passed in Retool
What was our Q3 profit?	45,000	Passed in Retool
What was our Q4 profit margin?	34.67%	Passed in Retool
Show profit margins by quarter	30.00%, 31.67%, 33.33%, 34.67%	Passed in notebook
Why did our European margins drop last quarter?	Report that sample margin increased from Q3 to Q4 by about 1.33 percentage points	Passed in Retool
Unsupported company-fact question	Unsupported status, no invented answer	Passed in Retool
Example-question set	Responses produced for all five dashboard examples	User-reported as working

Note: The margin comparison reports that the sample margin rose from 33.33% in Q3 to 34.67% in Q4, a rise of approximately 1.33 percentage points. This comparison does not establish the cause of any real-world margin movement.

10. Example API Response

```
{
  "success": true,
  "message": "Analysis calculated from sample demonstration data.",
  "answer": "Q3 sample revenue: 135,000",
  "value": 135000,
  "data_source": "Sample demonstration data",
  "metadata": {"status": "approved_demo"}
}
```

11. How to Run the Prototype

7. Open the Week 5 notebook in the environment used for development.
8. Run the cells that define the sample DataFrame and the metricmind_query function.
9. Start the FastAPI service on the configured port (8001).
10. If Retool connects through ngrok, start the tunnel and verify the configured backend URL.
11. Open the Retool dashboard and submit one of the supported example questions.
12. Inspect the answer, API response, and representative SQL panels.

The notebook kernel, API server, and tunnel may need to remain active during a live demonstration, depending on the deployment configuration.

12. Future Enhancements

- Connect the semantic layer to a real analytical database or warehouse.
- Execute validated SQL against a controlled database instead of showing illustrative SQL only.
- Implement centrally defined metrics, dimensions, joins, time periods, and regional filters.
- Expand the dataset to include multiple regions and detailed cost categories for meaningful breakdowns.
- Add stronger query validation, row limits, access controls, logging, and audit trails.
- Add automated unit and integration tests for supported and unsupported questions.
- Improve error handling, deployment reliability, and documentation.

13. Conclusion

MetricMind demonstrates a conversational BI workflow in which users ask selected business questions and receive structured answers through a Retool dashboard backed by a FastAPI service. The prototype successfully demonstrates quarterly revenue, profit, and margin calculations using sample data, along with basic unsupported-question handling and transparency displays. It provides a foundation for further work on semantic definitions, live database execution, and production-grade governance. Its current scope and limitations are documented so that the results are interpreted accurately.